SET-D

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B.SC.[ECS] - II Sem III Practical Examination, October 2026

VSC1: Based on DSC1-3 and DSC1-4

Date & Time: Max Marks: 30

Instructions:

I) Attempt any **one** question from each section.

II) Each question carries **10 marks.**

III) The **Journal** and **Viva-Voce** carry **5 marks.**

Section-I: Based on DSC1-3 (Data Structures using C++)

Q.1 Write a program to implement all queue operations using a linked list.

Q.2 Write a program to sort array elements in descending order using bubble sort.

Section-II: Based on DSC1-4 (Database Management Systems)

Q.3 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

Field Name	Datatype	Size	Description
sno	Varchar2	6	Primary Key
sname	Varchar2	20	Salesman name
address	Varchar2	30	Residential address
address1	Varchar2	30	Permanent address
pincode	number	6	Pin code
dob	Date		Date of birth
state	char	20	State
doj	date		Date of joining must be greater than Date of birth
department	Char	20	department name
salary	number	9,2	Salary of salesman

a) Change the column name address1 to permanent address.

b) List of salesmen whose salary is in order of Highest and lowest.

c) Display the average salary of salesmen.

d) Display the list of salesmen whose salary is maximum, minimum, and average salary department-wise.

e) Display how many salesmen are working in the computer department.

Q.4 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

A] Table Structure of Account_Master

Field Name	Datatype	Size	Description
Acc_no	Varchar2	6	Primary Key
Acc_name	Varchar2	20	Account holder's name
Gender	Char	1	Either M or F
Op_Date	Date		The date on which the Account is opened
Chq_fac	Char	1	Cheque facility status set to default 'N'
Cl_Date	Date		The date on which the Account is closed

B] Table Structure of Account_Transaction

Field Name	Datatype	Size	Description
Acc_no	Varchar2	6	Foreign Key
Tr_amount	Number	7,2	Transaction Amount
Tr_date	Date		Transaction Date
Chq_no	Varchar2	10	Cheque Number
Tr_type	Char	1	Receipt or Withdrawal

a) Display all the details of account holders in ascending order of Op_date.

b) Display the details of account holders who have a cheque facility and a balance >4000.

c) Display the details of account holders whose balance is between 4000 and 9000.

d) Display the name and number of female account holders.

e) Display the details of account holders whose 2^{nd} char of name is 'a'.